

GSERM - St. Gallen 2025

Analyzing Panel Data

June 26, 2026

Start with:

$$Y_i^* = \mathbf{X}_i\beta + u_i$$

$$Y_i = 0 \text{ if } Y_i^* < 0$$

$$Y_i = 1 \text{ if } Y_i^* \geq 0$$

So:

$$\begin{aligned} \Pr(Y_i = 1) &= \Pr(Y_i^* \geq 0) \\ &= \Pr(\mathbf{X}_i\beta + u_i \geq 0) \\ &= \Pr(u_i \geq -\mathbf{X}_i\beta) \\ &= \Pr(u_i \leq \mathbf{X}_i\beta) \\ &= \int_{-\infty}^{\mathbf{X}_i\beta} f(u) du \end{aligned}$$

“Standard logistic” PDF:

$$\Pr(u) \equiv \lambda(u) = \frac{\exp(u)}{[1 + \exp(u)]^2}$$

CDF:

$$\begin{aligned}\Lambda(u) &= \int \lambda(u) du \\ &= \frac{\exp(u)}{1 + \exp(u)} \\ &= \frac{1}{1 + \exp(-u)}\end{aligned}$$

Logistic $\rightarrow$ “Logit”

$$\begin{aligned}\Pr(Y_i = 1) &= \Pr(Y_i^* > 0) \\ &= \Pr(u_i \leq \mathbf{X}_i\boldsymbol{\beta}) \\ &= \Lambda(\mathbf{X}_i\boldsymbol{\beta}) \\ &= \frac{\exp(\mathbf{X}_i\boldsymbol{\beta})}{1 + \exp(\mathbf{X}_i\boldsymbol{\beta})}\end{aligned}$$

$$\text{(equivalently)} = \frac{1}{1 + \exp(-\mathbf{X}_i\boldsymbol{\beta})}$$

$$L_i = \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right)^{Y_i} \left[1 - \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) \right]^{1-Y_i}$$

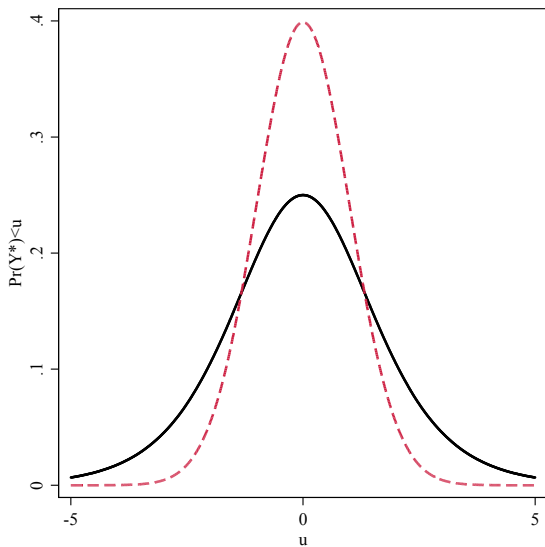
$$L = \prod_{i=1}^N \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right)^{Y_i} \left[1 - \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) \right]^{1-Y_i}$$

$$\begin{aligned} \ln L &= \sum_{i=1}^N Y_i \ln \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) + \\ &\quad (1 - Y_i) \ln \left[1 - \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) \right] \end{aligned}$$

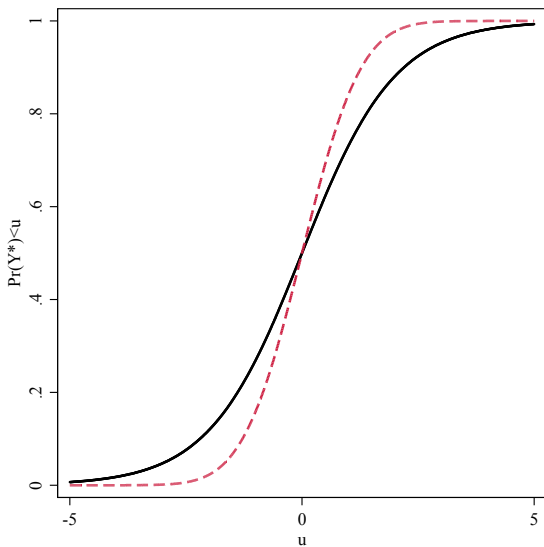
$$\Pr(u) \equiv \phi(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right)$$

$$\Phi(u) = \int_{-\infty}^u \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du$$

Standard Normal and Logistic PDFs



Standard Normal and Logistic CDFs



$$\begin{aligned}\Pr(Y_i = 1) &= \Phi(\mathbf{X}_i\beta) \\ &= \int_{-\infty}^{\mathbf{X}_i\beta} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\mathbf{X}_i\beta)^2}{2}\right) d\mathbf{X}_i\beta\end{aligned}$$

$$L = \prod_{i=1}^N [\Phi(\mathbf{X}_i\beta)]^{Y_i} [1 - \Phi(\mathbf{X}_i\beta)]^{(1-Y_i)}$$

$$\ln L = \sum_{i=1}^N Y_i \ln \Phi(\mathbf{X}_i\beta) + (1 - Y_i) \ln [1 - \Phi(\mathbf{X}_i\beta)]$$

One-way unit effects:

$$Y_{it} = f(\mathbf{X}_{it}\beta + \alpha_i + u_{it})$$

So, think about logit first:

$$\Pr(Y_{it} = 1) = \frac{\exp(\mathbf{X}_{it}\beta + \alpha_i)}{1 + \exp(\mathbf{X}_{it}\beta + \alpha_i)} \equiv \Lambda(\mathbf{X}_{it}\beta + \alpha_i)$$

Incidental Parameters:

- Nonlinearity $\rightarrow$ inconsistency in both $\hat{\alpha}$ s and $\hat{\beta}$.
- Anderson's *unconditional* estimator:

$$L^U = \prod_{i=1}^N \prod_{t=1}^T \Lambda(\mathbf{X}_{it} + \alpha_i)^{Y_{it}} [1 - \Lambda(\mathbf{X}_{it} + \alpha_i)]^{1-Y_{it}}$$

- Chamberlain's *conditional* estimator:

$$L^C = \prod_{i=1}^N \Pr \left(Y_{i1} = y_{i1}, Y_{i2} = y_{i2}, \dots, Y_{iT} = y_{iT} \mid \sum_{t=1}^T Y_{it} \right)$$

Fixed-Effects (continued)

Intuition: Suppose we have $T = 2$. That means that:

- $\Pr(Y_{i1} = 0 \text{ and } Y_{i2} = 0 \mid \sum_T Y_{it} = 0) = 1.0$
- $\Pr(Y_{i1} = 1 \text{ and } Y_{i2} = 1 \mid \sum_T Y_{it} = 2) = 1.0$

and:

$$\Pr\left(Y_{i1} = 0 \text{ and } Y_{i2} = 1 \mid \sum_T Y_{it} = 1\right) = \frac{\Pr(0, 1)}{\Pr(0, 1) + \Pr(1, 0)}$$

with a similar statement for $\Pr(Y_{i1} = 1 \text{ and } Y_{i2} = 0 \mid \sum_T Y_{it} = 1)$.

The Point:

$\sum_{t=1}^T Y_{it}$ is a sufficient statistic for α_i , so conditioning on it $\equiv$ “fixed effects.”

Things to bear in mind:

- Fixed effects = no estimates for β_b
- Interpretation: per logit, but $|\hat{\alpha}_j$.
- Everything above is for **logit**...
 - For FE probit, there is no conditional model
 - Unconditional / “brute force” FE probit is biased (see [here](#) and [here](#))
- BTSCS in international relations: [Green et al. \(2001\)](#) vs. [Beck & Katz \(2001\)](#) (“Dirty Pool” debate)

Model is:

$$\begin{aligned} Y_{it}^* &= \mathbf{X}_{it}\beta + u_{it} \\ Y_{it} &= 0 \text{ if } Y_{it}^* \leq 0 ; \\ &= 1 \text{ if } Y_{it}^* > 0 \end{aligned}$$

with:

$$u_{it} = \alpha_i + \eta_{it}$$

with $\eta_{it} \sim \text{i.i.d. } N(0,1)$, and $\alpha_i \sim N(0, \sigma_\alpha^2)$. This implies:

$$\text{Var}(u_{it}) = 1 + \sigma_\alpha^2$$

and so:

$$\text{Corr}(u_{it}, u_{is}, t \neq s) \equiv \rho = \frac{\sigma_\alpha^2}{1 + \sigma_\alpha^2}$$

which means that we can write $\sigma_\alpha^2 = \left(\frac{\rho}{1-\rho} \right)$.

Probit:

$$\begin{aligned} L_i &= \text{Prob}(Y_{i1} = y_{i1}, Y_{i2} = y_{i2}, \dots Y_{iT} = y_{iT}) \\ &= \int_{-\infty}^{X_{i1}\beta} \int_{-\infty}^{X_{i2}\beta} \dots \int_{-\infty}^{X_{iT}\beta} \phi(u_{i1}, u_{i2} \dots u_{iT}) du_{iT} \dots du_{i2} du_{i1} \end{aligned}$$

Logit:

$$\begin{aligned} L_i &= \text{Prob}(Y_{i1} = y_{i1}, Y_{i2} = y_{i2}, \dots Y_{iT} = y_{iT}) \\ &= \int_{-\infty}^{X_{i1}\beta} \int_{-\infty}^{X_{i2}\beta} \dots \int_{-\infty}^{X_{iT}\beta} \lambda(u_{i1}, u_{i2} \dots u_{iT}) du_{iT} \dots du_{i2} du_{i1} \end{aligned}$$

Solution?

$$\phi(u_{i1}, u_{i2}, \dots u_{iT}) = \int_{-\infty}^{\infty} \phi(u_{i1}, u_{i2}, \dots u_{iT} \mid \alpha_i) \phi(\alpha_i) d\alpha_i$$

Points:

- $\hat{\rho}$ = proportion of the variance due to the α_i s.
- Implementation: Gauss-Hermite quadrature or MCMC.
- Best with N large and T small.
- Critically requires $\text{Cov}(\mathbf{X}, \alpha) = 0$ (see notes re: Chamberlain's CRE Estimator).

R

- `pglm` (panel GLMs) (maximum likelihood + quadrature)
- `bife` (fixed-effects logit / probit only)
- `glmer` (general mixed-effects models, including RE)
- `glmmML` (via Gauss-Hermite quadrature)
- `MCMCpack` (`MCMChlogit`)
- Various user-generated functions (e.g., [here](#)).
- Interpretation via `modelsummary` and `marginaleffects`

Stata

- `xtprobit`, `xtlogit`, `xtcloglog`
- Plus `xttrans` (transition probabilities), `quadchk` (quadrature checking), `xtrho` / `xtrhoi` (estimation of within-unit covariances)

Example: WDI “Plus”

Data from the WDI, plus POLITY and the UCDP:

- IS03 - The country's International Standards Organization (ISO) three-letter identification code.
- Year - The year that row of data applies to (1960=1).
- CivilWar - Civil conflict indicator: 1 if there was a civil conflict in that country in that year; 0 otherwise. From the UCDP.
- OnsetCount - The sum of new conflict episodes in that country / year. From UCDP.
- LandArea - Land area (sq. km).
- PopMillions - Population (in millions).
- PopGrowth - Population Growth (percent).
- UrbanPopulation - Urban Population (percent of total).
- GDPPerCapita - GDP per capita (constant 2010 \$US).
- GDPPerCapGrowth - GDP Per Capita Growth (percent annual).
- PostColdWar - 1 if Year > 1989, 0 otherwise.
- POLITY - The POLITY score of democracy/autocracy. Scaled so that 0 = most autocratic, 10 = most democratic.

$N = 215$, $\bar{T} = 64$, NT varies (due to missingness).

Model:

$$\begin{aligned} \text{Civil War}_{it} = & f[\beta_0 + \beta_1 \ln(\text{Land Area}_{it}) + \beta_2 \ln(\text{Population}_{it}) + \\ & + \beta_3 \text{Urban Population}_{it} + \beta_4 \ln(\text{GDP}_{it}) + \beta_5 \text{GDP Growth}_{it} + \\ & + \beta_6 \text{Post-Cold War}_{it} + \beta_7 \text{POLITY}_{it} + \beta_5 \text{POLITY}_{it}^2 + u_{it}] \end{aligned}$$

```
> describe(DF,skew=FALSE)
```

	vars	n	mean	sd	median	min	max	range	se
IS03*	1	14252	108.47	62.35	108.00	1.00	216.0	215.00	0.52
Year*	2	14252	33.49	19.05	33.00	1.00	66.0	65.00	0.16
YearNumeric	3	14190	1992.50	19.05	1992.50	1960.00	2025.0	65.00	0.16
country*	4	14190	108.00	62.07	108.00	1.00	215.0	214.00	0.52
CivilWar	5	9378	0.13	0.34	0.00	0.00	1.0	1.00	0.00
OnsetCount	6	9378	0.05	0.24	0.00	0.00	4.0	4.00	0.00
LandArea	7	12288	596587.32	1645011.14	100540.00	2.03	16389950.0	16389947.97	14839.81
PopMillions	8	13945	25.54	106.13	4.33	0.00	1450.9	1450.93	0.90
UrbanPopulation	9	13975	52.21	25.80	52.12	1.84	100.0	98.16	0.22
GDPPerCapita	10	11307	12420.38	19440.07	3997.77	122.68	247170.2	247047.54	182.82
GDPPerCapGrowth	11	11184	1.91	6.54	2.10	-64.42	140.5	204.91	0.06
PostColdWar	12	14190	0.55	0.50	1.00	0.00	1.0	1.00	0.00
POLITY	13	8310	5.60	3.68	6.50	0.00	10.0	10.00	0.04
POLITYSquared	14	8310	44.95	40.04	42.25	0.00	100.0	100.00	0.44

Variable	Dim	Mean	SD	Min	Max	Observations
YearNumeric	overall	1992.5	19.051	1960	2025	N = 14190
	between		0	1992.5	1992.5	n = 215
	within		19.051	1960	2025	T = 66
CivilWar	overall	0.133	0.339	0	1	N = 9378
	between		0.222	0	1	n = 171
	within		0.253	-0.787	1.117	T = 54.842
OnsetCount	overall	0.049	0.242	0	4	N = 9378
	between		0.083	0	0.597	n = 171
	within		0.227	-0.548	3.92	T = 54.842
LandArea	overall	596587.319	1645011.136	2.027	16389950	N = 12288
	between		1754315.837	2.03	16379186.875	n = 215
	within		5522.286	428120.652	651733.338	T = 57.153
PopMillions	overall	25.538	106.126	0.003	1450.936	N = 13945
	between		102.116	0.008	1113.738	n = 215
	within		29.352	-458.817	556.128	T = 64.86
UrbanPopulation	overall	52.209	25.796	1.839	100	N = 13975
	between		23.952	8.128	100	n = 215
	within		9.715	0.976	89.053	T = 65
GDPPerCapita	overall	12420.376	19440.071	122.679	247170.22	N = 11307
	between		21065.471	286.863	167187.157	n = 211
	within		7465.265	-40583.893	133748.603	T = 53.588
GDPPerCapGrowth	overall	1.908	6.541	-64.424	140.491	N = 11184
	between		1.692	-7.984	7.697	n = 212
	within		6.362	-66.273	134.702	T = 52.755
PostColdWar	overall	0.545	0.498	0	1	N = 14190
	between		0	0.545	0.545	n = 215
	within		0.498	0	1	T = 66
POLITY	overall	5.604	3.681	0	10	N = 8310
	between		2.967	0	10	n = 163
	within		2.219	-1.421	12.312	T = 50.982
POLITYSquared	overall	44.95	40.035	0	100	N = 8310
	between		33.014	0	100	n = 163
	within		22.853	-30.427	115.645	T = 50.982

Pooled Logit

```
> Logit<-glm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+  
+           GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared,data=DF,family="binomial")
```

```
> summary(Logit)
```

Call:

```
glm(formula = CivilWar ~ log(LandArea) + log(PopMillions) + UrbanPopulation +  
    log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar + POLITY +  
    POLITYSquared, family = "binomial", data = DF)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.20188	0.49219	-0.41	0.68169
log(LandArea)	-0.11586	0.03027	-3.83	0.00013 ***
log(PopMillions)	0.69996	0.03572	19.59	< 2e-16 ***
UrbanPopulation	0.01438	0.00294	4.90	9.6e-07 ***
log(GDPPerCapita)	-0.44336	0.05406	-8.20	2.4e-16 ***
GDPPerCapGrowth	-0.03123	0.00602	-5.18	2.2e-07 ***
PostColdWar	-0.47639	0.08337	-5.71	1.1e-08 ***
POLITY	0.72826	0.06160	11.82	< 2e-16 ***
POLITYSquared	-0.06950	0.00578	-12.02	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 6010.2 on 7504 degrees of freedom
Residual deviance: 4929.1 on 7496 degrees of freedom
(6747 observations deleted due to missingness)
AIC: 4947

Number of Fisher Scoring iterations: 6

Fixed Effects

```
> FELogit<-bife(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+               GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|ISO3,data=DF,model="logit")
```

```
> summary(FELogit)
binomial - logit link
```

```
CivilWar ~ log(LandArea) + log(PopMillions) + UrbanPopulation +
  log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar + POLITY +
  POLITYSquared | ISO3
```

Estimates:

	Estimate	Std. error	z	value	Pr(> z)
log(LandArea)	-13.78997	9.31125	-1.48	0.139	
log(PopMillions)	0.52131	0.26401	1.97	0.048	*
UrbanPopulation	0.01249	0.00903	1.38	0.167	
log(GDPPerCapita)	-0.39393	0.15951	-2.47	0.014	*
GDPPerCapGrowth	-0.04406	0.00780	-5.65	1.6e-08	***
PostColdWar	-0.11913	0.17288	-0.69	0.491	
POLITY	0.70726	0.09291	7.61	2.7e-14	***
POLITYSquared	-0.07424	0.00875	-8.49	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
residual deviance= 2952,
null deviance= 4547,
n= 4214, N= 84
```

```
( 6747 observation(s) deleted due to missingness )
( 3291 observation(s) deleted due to perfect classification )
```

Number of Fisher Scoring Iterations: 6

Average individual fixed effect= 172.2

Alternative Fixed Effects (using feglm)

```
> FELogit2<-feglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+               GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|ISO3,data=DF,family="binomial")
NOTES: 6,747 observations removed because of NA values (LHS: 4,874, RHS: 6,747).
       76 fixed-effects (3,291 observations) removed because of only 0 or only 1 outcomes
       or singletons.
```

```
> summary(FELogit2)
```

GLM estimation, family = binomial, Dep. Var.: CivilWar

Observations: 4,214

Fixed-effects: ISO3: 84

Standard-errors: IID

	Estimate	Std. Error	z value	Pr(> z)
log(LandArea)	-13.78995	9.413465	-1.4649	1.4294e-01
log(PopMillions)	0.52132	0.266912	1.9531	5.0803e-02 .
UrbanPopulation	0.01249	0.009131	1.3678	1.7137e-01
log(GDPPerCapita)	-0.39393	0.161257	-2.4429	1.4571e-02 *
GDPPerCapGrowth	-0.04406	0.007888	-5.5864	2.3182e-08 ***
PostColdWar	-0.11913	0.174782	-0.6816	4.9548e-01
POLITY	0.70726	0.093930	7.5297	5.0855e-14 ***
POLITYSquared	-0.07424	0.008844	-8.3949	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1,475.8 Adj. Pseudo R2: 0.310836

BIC: 3,719.4 Squared Cor.: 0.388961

Random Effects

```
> RELogit<-pglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+               GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared|IS03,data=DF,family=binomial
+               effect="individual",model="random",method="bfgs")
```

```
> summary(RELogit)
```

Maximum Likelihood estimation

BFGS maximization, 206 iterations

Return code 0: successful convergence

Log-Likelihood: -1689

10 free parameters

Estimates:

	Estimate	Std. error	t value	Pr(> t)
(Intercept)	-0.15113	1.04945	-0.14	0.89
log(LandArea)	-0.05012	0.05874	-0.85	0.39
log(PopMillions)	0.97490	0.07746	12.59	< 2e-16 ***
UrbanPopulation	0.00333	0.00508	0.66	0.51
log(GDPPerCapita)	-0.67386	0.11253	-5.99	0.0000000021 ***
GDPPerCapGrowth	-0.04290	0.00755	-5.69	0.0000000130 ***
PostColdWar	-0.16196	0.12256	-1.32	0.19
POLITY	0.71733	0.08325	8.62	< 2e-16 ***
POLITYSquared	-0.07636	0.00808	-9.45	< 2e-16 ***
sigma	1.98996	0.09384	21.21	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Models of Civil War

	Logit	FE Logit	FES+Robust	RE Logit
Intercept	-0.20 (0.49)			-0.15 (1.05)
ln(Land Area)	-0.12* (0.03)	-13.79 (9.31)	-13.79 (9.41)	-0.05 (0.06)
ln(Population)	0.70* (0.04)	0.52* (0.26)	0.52 (0.27)	0.97* (0.08)
Urban Population	0.01* (0.00)	0.01 (0.01)	0.01 (0.01)	0.00 (0.01)
ln(GDP Per Capita)	-0.44* (0.05)	-0.39* (0.16)	-0.39* (0.16)	-0.67* (0.11)
GDP Growth	-0.03* (0.01)	-0.04* (0.01)	-0.04* (0.01)	-0.04* (0.01)
Post-Cold War	-0.48* (0.08)	-0.12 (0.17)	-0.12 (0.17)	-0.16 (0.12)
POLITY	0.73* (0.06)	0.71* (0.09)	0.71* (0.09)	0.72* (0.08)
POLITY Squared	-0.07* (0.01)	-0.07* (0.01)	-0.07* (0.01)	-0.08* (0.01)
Estimated Sigma				1.99* (0.09)
AIC	4947.13			3398.14
BIC	5009.44			
Log Likelihood	-2464.57	-1475.78	-1475.78	-1689.07
Deviance	4929.13	2951.56	2951.56	
Num. obs.	7505	4214	4214	
Num. groups: ISO3			84	

* $p < 0.05$

Models For Event Counts

Properties:

- Discrete / integer-values
- Non-negative
- “Cumulative”

Motivation:

$$\text{Arrival Rate} = \lambda$$

$$\Pr(\text{Event})_{t,t+h} = \lambda h$$

$$\Pr(\text{No Event})_{t,t+h} = 1 - \lambda h$$

$$\begin{aligned}\Pr(Y_t = y) &= \frac{\exp(-\lambda h) \lambda h^y}{y!} \\ &= \frac{\exp(-\lambda) \lambda^y}{y!}\end{aligned}$$

Poisson: Assumptions and Motivations

Three key assumptions:

- No Simultaneous Events
- Constant Arrival Rate
- Independent Event Arrivals

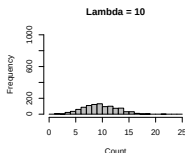
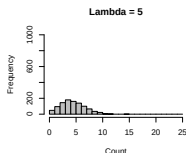
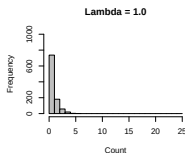
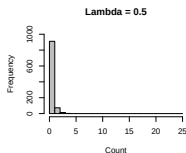
Another motivation: For M independent Bernoulli trials with (sufficiently small) probability of success π and where $M\pi \equiv \lambda > 0$,

$$\begin{aligned}\Pr(Y_i = y) &= \lim_{M \rightarrow \infty} \left[\binom{M}{y} \left(\frac{\lambda}{M}\right)^y \left(1 - \frac{\lambda}{M}\right)^{M-y} \right] \\ &= \frac{\lambda^y \exp(-\lambda)}{y!}\end{aligned}$$

Poisson: Characteristics

Characteristics:

- Discrete
- $E(Y) = \text{Var}(Y) = \lambda$
- Is not preserved under affine transformations...
- For $X \sim \text{Poisson}(\lambda_X)$ and $Y \sim \text{Poisson}(\lambda_Y)$, $Z = X + Y \sim \text{Poisson}(\lambda_X + \lambda_Y)$
iff X and Y are independent but
- ...same is not true for differences.
- $\lambda \rightarrow \infty \iff Y \sim N$



Suppose

$$E(Y_i) \equiv \lambda_i = \exp(\mathbf{X}_i\beta)$$

then

$$\Pr(Y_i = y | \mathbf{X}_i, \beta) = \frac{\exp[-\exp(\mathbf{X}_i\beta)][\exp(\mathbf{X}_i\beta)]^y}{y!}$$

with likelihood:

$$L = \prod_{i=1}^N \frac{\exp[-\exp(\mathbf{X}_i\beta)][\exp(\mathbf{X}_i\beta)]^{Y_i}}{Y_i!}$$

and log-likelihood:

$$\ln L = \sum_{i=1}^N [-\exp(\mathbf{X}_i\beta) + Y_i\mathbf{X}_i\beta - \ln(Y_i!)]$$

Event Counts: Unit Effects

The Poisson model:

$$Y_{it} \sim \text{Poisson}(\mu_{it} = \alpha_i \lambda_{it})$$

with $\lambda_{it} = \exp(\mathbf{X}_{it}\beta)$ implies:

$$\begin{aligned} E(Y_{it} \mid \mathbf{X}_{it}, \alpha_i) &= \mu_{it} \\ &= \alpha_i \exp(\mathbf{X}_{it}\beta) \\ &= \exp(\delta_i + \mathbf{X}_{it}\beta) \end{aligned}$$

where $\delta_i = \ln(\alpha_i)$.

Fixed-Effects Poisson:

- ...has no “incidental parameters” problem (see e.g. Cameron and Trivedi, pp. 281-2)
- This means “brute force” approach works
- Fitted via `glmmML` in R, `xtpoisson` (and `xtnbreg`) in Stata

The Poisson with random effects is:

$$\begin{aligned}\Pr(Y_{i1} = y_{i1}, \dots, Y_{iT} = y_{iT}) &= \int_0^\infty \Pr(Y_{i1} = y_{i1}, \dots, Y_{iT} = y_{iT}) f(\alpha_i) d\alpha_i \\ &= \int_0^\infty \left[\prod_{t=1}^T \Pr(Y_{it} | \alpha_i) \right] f(\alpha_i) d\alpha_i\end{aligned}$$

- Simplest to assume $\alpha_i \sim \Gamma(\theta)$
- Yields a model with $E(Y_{it}) = \lambda_{it}$ and $\text{Var}(Y_{it}) = \lambda_{it} + \frac{\lambda_{it}^2}{\theta}$
- Via `glmmML` or `glmer` in R, or `xtpois`, `re` in Stata
- $\exists$ random effects negative binomial too...

R:

- Poisson + negative binomial (FE, RE) = `pglm`
- Poisson + negative binomial (+ logit) (FE, RE) = `fixest`
- Poisson (random effects) = `glmmML` in **glmmML** or `glmer` in **lme4**
- Poisson (fixed effects) = `glmmML` or “brute force”

Stata:

- Tobit = `xttobit` (re only)
- Poisson / negative binomial = `xtpoisson`, `xtnbreg` (both with `fe`, `re` options)

Conflict Onsets: Pooled Poisson

```
> xtabs(~DF$OnsetCount)
DF$OnsetCount
  0    1    2    3    4
8966 374   30    7    1

> Poisson<-glm(OnsetCount~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+             GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared,data=DF,family="poisson")

> summary(Poisson)

Call:
glm(formula = OnsetCount ~ log(LandArea) + log(PopMillions) +
    UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar +
    POLITY + POLITYSquared, family = "poisson", data = DF)

Coefficients:
              Estimate Std. Error z value      Pr(>|z|)
(Intercept)   -2.13110    0.69036   -3.09      0.002 **
log(LandArea)    0.03343    0.04561    0.73      0.464
log(PopMillions) 0.44243    0.04465    9.91    < 2e-16 ***
UrbanPopulation  0.00700    0.00423    1.66      0.098 .
log(GDPPerCapita) -0.42350    0.07363   -5.75 0.0000000089 ***
GDPPerCapGrowth -0.03468    0.00642   -5.40 0.0000000672 ***
PostColdWar      0.20054    0.11936    1.68      0.093 .
POLITY           0.39283    0.08484    4.63 0.0000036559 ***
POLITYSquared    -0.04250    0.00814   -5.22 0.0000001747 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 2456.1  on 7504  degrees of freedom
Residual deviance: 2018.3  on 7496  degrees of freedom
(6747 observations deleted due to missingness)
AIC: 2783

Number of Fisher Scoring iterations: 6
```

Fixed Effects Poisson

```
> FEpoisson<-pglm(OnsetCount~log(LandArea)+log(PopMillions)+UrbanPopulation+
+               log(GDPPerCapita)+GDPPerCapGrowth+PostColdWar+POLITY+
+               POLITYSquared,data=DF,family="poisson",effect="individual",
+               model="within")

> summary(FEpoisson)
-----
Maximum Likelihood estimation
Newton-Raphson maximisation, 4 iterations
Return code 8: successive function values within relative tolerance limit (reltol)
Log-Likelihood: -1049
8 free parameters
Estimates:

```

	Estimate	Std. error	t value	Pr(> t)
log(LandArea)	-12.88863	8.70860	-1.48	0.139
log(PopMillions)	0.45449	0.29549	1.54	0.124
UrbanPopulation	-0.02039	0.01103	-1.85	0.065 .
log(GDPPerCapita)	-0.19760	0.14019	-1.41	0.159
GDPPerCapGrowth	-0.03057	0.00657	-4.66	0.0000032 ***
PostColdWar	0.32337	0.19138	1.69	0.091 .
POLITY	0.44900	0.10463	4.29	0.0000178 ***
POLITYSquared	-0.04644	0.01007	-4.61	0.0000040 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
-----
```

Equivalent Fixed Effects Poisson (using feglm)

```
> FEpoisson2<-feglm(OnsetCount~log(LandArea)+log(PopMillions)+UrbanPopulation+
+                   log(GDPPerCapita)+GDPPerCapGrowth+PostColdWar+POLITY+
+                   POLITYSquared|IS03,data=DF,family="poisson")
NOTES: 6,747 observations removed because of NA values (LHS: 4,874, RHS: 6,747).
       64 fixed-effects (2,725 observations) removed because of only 0 outcomes
       or singletons.
```

```
> summary(FEpoisson2,cluster="IS03")
GLM estimation, family = poisson, Dep. Var.: OnsetCount
Observations: 4,780
Fixed-effects: IS03: 96
Standard-errors: Clustered (IS03)
```

	Estimate	Std. Error	z value	Pr(> z)
log(LandArea)	-12.88872	8.847185	-1.457	0.14516733025
log(PopMillions)	0.45449	0.351920	1.291	0.19654509850
UrbanPopulation	-0.02039	0.018512	-1.101	0.27079576689
log(GDPPerCapita)	-0.19760	0.187396	-1.054	0.29168499578
GDPPerCapGrowth	-0.03057	0.006223	-4.913	0.00000089749 ***
PostColdWar	0.32337	0.267247	1.210	0.22627785185
POLITY	0.44900	0.132077	3.400	0.00067497219 ***
POLITYSquared	-0.04644	0.012502	-3.715	0.00020338804 ***

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
Log-Likelihood: -1,189.7    Adj. Pseudo R2: 0.08875
               BIC: 3,260.5    Squared Cor.: 0.15562
```

Random Effects Poisson

```
> REPoisson<-pglm(OnsetCount~log(LandArea)+log(PopMillions)+UrbanPopulation+
+               log(GDPPerCapita)+GDPPerCapGrowth+PostColdWar+POLITY+
+               POLITYSquared,data=DF,family="poisson",effect="individual",
+               model="random")
```

```
> summary(REPoisson)
```

Maximum Likelihood estimation

Newton-Raphson maximisation, 4 iterations

Return code 8: successive function values within relative tolerance limit (reltol)

Log-Likelihood: -1330

10 free parameters

Estimates:

	Estimate	Std. error	t value	Pr(> t)
(Intercept)	-2.73263	1.02006	-2.68	0.0074 **
log(LandArea)	0.00500	0.07157	0.07	0.9444
log(PopMillions)	0.45487	0.07967	5.71	0.0000000113 ***
UrbanPopulation	-0.00128	0.00571	-0.22	0.8222
log(GDPPerCapita)	-0.26273	0.09564	-2.75	0.0060 **
GDPPerCapGrowth	-0.03212	0.00654	-4.91	0.0000008925 ***
PostColdWar	0.21314	0.12872	1.66	0.0977 .
POLITY	0.48579	0.09447	5.14	0.0000002713 ***
POLITYSquared	-0.05308	0.00909	-5.84	0.0000000053 ***
sigma	1.67732	0.39765	4.22	0.0000246439 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Negative Binomial: Fixed Effects

```
> FENegBin<-pglm(OnsetCount~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+
+               GDPPerCapGrowth+PostColdWar+POLITY+POLITYSquared,data=DF,family="negbin",
+               effect="individual",model="within")
```

```
> summary(FENegBin)
```

Maximum Likelihood estimation

Newton-Raphson maximisation, 9 iterations

Return code 1: gradient close to zero (gradtol)

Log-Likelihood: -1050

9 free parameters

Estimates:

	Estimate	Std. error	t value	Pr(> t)
(Intercept)	2.41088	6.47804	0.37	0.710
log(LandArea)	-0.00932	0.53588	-0.02	0.986
log(PopMillions)	0.50204	0.27955	1.80	0.073 .
UrbanPopulation	-0.02395	0.01061	-2.26	0.024 *
log(GDPPerCapita)	-0.20478	0.13976	-1.47	0.143
GDPPerCapGrowth	-0.03027	0.00662	-4.57	0.0000048 ***
PostColdWar	0.35825	0.19016	1.88	0.060 .
POLITY	0.44770	0.10567	4.24	0.0000227 ***
POLITYSquared	-0.04705	0.01016	-4.63	0.0000036 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Models of Civil War Onset Counts

	Poisson	FE Poisson	RE Poisson	FE Neg. Bin.
Intercept	-2.13* (0.69)		-2.73* (1.02)	2.41 (6.48)
ln(Land Area)	0.03 (0.05)	-12.89 (8.71)	0.00 (0.07)	-0.01 (0.54)
ln(Population)	0.44* (0.04)	0.45 (0.30)	0.45* (0.08)	0.50 (0.28)
Urban Population	0.01 (0.00)	-0.02 (0.01)	-0.00 (0.01)	-0.02* (0.01)
ln(GDP Per Capita)	-0.42* (0.07)	-0.20 (0.14)	-0.26* (0.10)	-0.20 (0.14)
GDP Growth	-0.03* (0.01)	-0.03* (0.01)	-0.03* (0.01)	-0.03* (0.01)
Post-Cold War	0.20 (0.12)	0.32 (0.19)	0.21 (0.13)	0.36 (0.19)
POLITY	0.39* (0.08)	0.45* (0.10)	0.49* (0.09)	0.45* (0.11)
POLITY Squared	-0.04* (0.01)	-0.05* (0.01)	-0.05* (0.01)	-0.05* (0.01)
Estimated Sigma			1.68* (0.40)	
AIC	2783.26	2114.61	2680.02	2118.52
BIC	2845.57			
Log Likelihood	-1382.63	-1049.30	-1330.01	-1050.26
Deviance	2018.31			
Num. obs.	7505			

* $p < 0.05$

Wrap-Up: Some Useful Packages

- `pglm`
 - Workhorse package for panel (FE, RE, BE) GLMs
 - Binary + ordered logit/probit, Poisson / negative binomial
 - Discussed + used extensively in Croissant and Millo (2018) *Panel Data Econometrics with R*
 - The one thing it won't (apparently) do is fixed-effects, binary-response models...
- `fixest`
 - Fast / efficient fitting of FE models
 - Fits linear models, logit, Poisson, and negative binomial
 - Includes easy coefficient plots & tables; simple multi-threading; built-in "robust" S.E.s
- `alpaca`
 - Fast / efficient fitting of GLMs with high-dimensional fixed effects
 - *Includes bias correction for incidental parameters after binary-response models*
 - Also includes useful panel data simulation routines + average partial effects

Generalized Estimating Equations

Linear-normal model is:

$$Y_i = \mu_i + u_i$$

with:

$$\mu_i = \mathbf{X}_i\boldsymbol{\beta}.$$

Generalize:

$$g(\mu_i) = \mathbf{X}_i\boldsymbol{\beta}$$

and:

$$Y_i \sim \text{i.i.d. } F[\mu_i, \mathbf{V}_i].$$

“Score” equations:

$$\mathbf{U}(\beta) = \sum_{i=1}^N \mathbf{D}_i' \mathbf{V}_i^{-1} [Y_i - \mu_i] = \mathbf{0}.$$

with:

- $\mathbf{D}_i = \frac{\partial \mu_i}{\partial \beta}$,
- $\mathbf{V}_i = \frac{h(\mu_i)}{\phi}$, and
- $(Y_i - \mu_i) \approx$ a “residual.”
- Known as “quasi-likelihood” (e.g. Wedderburn 1974 *Biometrika*).

Now suppose:

$$Y_{it} = \mu_{it} + u_{it}$$

where

- $i \in \{1, \dots, N\}$ are i.i.d. “units,”
- $t \in \{1, \dots, T\}$, $T > 1$ are “time points,”
- we want $g(\mu_{it}) = \mathbf{X}_{it}\beta$.

Key issue: Accounting for (conditional) dependence in Y over time.

Full joint distributions over T are hard. But...

Define:

$$\mathbf{R}_i(\boldsymbol{\alpha})_{T \times T} = \begin{pmatrix} 1.0 & \alpha_{12} & \cdots & \alpha_{1,T} \\ \alpha_{21} & 1.0 & \cdots & \alpha_{2,T} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{T,1} & \cdots & \alpha_{T,T-1} & 1.0 \end{pmatrix},$$

→ “working correlation” matrix.

- Completely defined by $\boldsymbol{\alpha}$,
- Structure specified by the analyst.

Liang and Zeger (1986): We can decompose the variance of Y_{it} as:

$$\mathbf{V}_i = \text{diag}(\mathbf{V}_i^{\frac{1}{2}}) \mathbf{R}_i(\boldsymbol{\alpha}) \text{diag}(\mathbf{V}_i^{\frac{1}{2}})$$

With a standard GLM assumption about the mean and variance, this is:

$$\mathbf{V}_i = \frac{(\mathbf{A}_i^{\frac{1}{2}}) \mathbf{R}_i(\boldsymbol{\alpha}) (\mathbf{A}_i^{\frac{1}{2}})}{\phi}$$

where

$$\mathbf{A}_i = \begin{pmatrix} h(\mu_{i1}) & 0 & \cdots & 0 \\ 0 & h(\mu_{i2}) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & h(\mu_{iT}) \end{pmatrix}$$

$\mathbf{V}_i = \text{Var}(Y_{it} | \mathbf{X}_{it}, \beta)$ has two parts:

- $\mathbf{A}_i = \text{unit-level}$ variation,
- $\mathbf{R}_i(\alpha) = \text{within-unit } \text{temporal}$ variation.

Specifying $\mathbf{R}_i(\alpha)$

Independent:

$$\mathbf{R}_i(\alpha) = \begin{pmatrix} 1.0 & 0 & \cdots & 0 \\ 0 & 1.0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & 1.0 \end{pmatrix}$$

- Assumes no within-unit temporal correlation.
- Equivalent to GLM on pooled data.

Exchangeable:

$$\mathbf{R}_i(\alpha) = \begin{pmatrix} 1.0 & \alpha & \cdots & \alpha \\ \alpha & 1.0 & \cdots & \alpha \\ \vdots & \vdots & \ddots & \vdots \\ \alpha & \cdots & \alpha & 1.0 \end{pmatrix}$$

- One free parameter in $\mathbf{R}_i(\alpha)$ ($\alpha_{ts} = \alpha \forall t \neq s$)
- Temporal correlation within units is constant across time points.
- Akin (in some respects) to a random-effects model...

Specifying $\mathbf{R}_i(\alpha)$

$AR(p)$ (e.g., $AR(1)$):
$$\mathbf{R}_i(\alpha) = \begin{pmatrix} 1.0 & \alpha & \alpha^2 & \cdots & \alpha^{T-1} \\ \alpha & 1.0 & \alpha & \cdots & \alpha^{T-2} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha^{T-1} & \cdots & \alpha^2 & \alpha & 1.0 \end{pmatrix}$$

- One free parameter in $\mathbf{R}_i(\alpha)$ ($\alpha_{ts} = \alpha^{|t-s|} \forall t \neq s$).
- Conditional within-unit correlation an exponential function of the lag.

$Stationary(p)$:
$$\mathbf{R}_i(\alpha) = \begin{pmatrix} 1.0 & \alpha_1 & \cdots & \alpha_p & 0 & \cdots & 0 \\ \alpha_1 & 1.0 & \alpha_1 & \cdots & \cdots & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & \cdots & 0 & \alpha_p & \cdots & \alpha_1 & 1.0 \end{pmatrix}$$

- AKA “banded,” or “ p -dependent.”
- $p \leq T - 1$ free parameters in $\mathbf{R}_i(\alpha)$.
- Conditional within-unit correlation an exponential function of the lag, up to lag p , and zero thereafter.

Specifying $\mathbf{R}_i(\alpha)$

Unstructured: $\mathbf{R}_i(\alpha) = \begin{pmatrix} 1.0 & \alpha_{12} & \alpha_{13} & \cdots & \alpha_{1,T-1} \\ \alpha_{12} & 1.0 & \alpha_{23} & \cdots & \alpha_{2,T-1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_{1,T-1} & \alpha_{2,T-1} & \cdots & \alpha_{T-1,T-1} & 1.0 \end{pmatrix}$

- $\frac{T(T-1)}{2}$ free parameters in $\mathbf{R}_i(\alpha)$.
- Conditional within-unit correlation is completely data-dependent.

Score equations:

$$\mathbf{U}_{GEE}(\boldsymbol{\beta}_{GEE}) = \sum_{i=1}^N \mathbf{D}_i' \left[\frac{(\mathbf{A}_i^{\frac{1}{2}}) \mathbf{R}_i(\boldsymbol{\alpha}) (\mathbf{A}_i^{\frac{1}{2}})}{\phi} \right]^{-1} [Y_i - \mu_i] = \mathbf{0}$$

Two-step estimation:

- For fixed values of $\boldsymbol{\alpha}_s$ and ϕ_s at iteration s , use Newton scoring to estimate $\hat{\boldsymbol{\beta}}_s$,
- Use $\hat{\boldsymbol{\beta}}_s$ to calculate standardized residuals $(Y_i - \hat{\mu}_i)_s$, from which consistent estimates of $\boldsymbol{\alpha}_{s+1}$ and ϕ_{s+1} can be estimated.

Liang & Zeger (1986):

$$\hat{\beta}_{GEE} \underset{N \rightarrow \infty}{\sim} \mathbf{N}(\beta, \Sigma).$$

For $\hat{\Sigma}$, two options:

$$\hat{\Sigma}_{\text{Model}} = N \left(\sum_{i=1}^N \hat{\mathbf{D}}_i' \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right)$$

$$\hat{\Sigma}_{\text{Robust}} = N \left(\sum_{i=1}^N \hat{\mathbf{D}}_i' \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right)^{-1} \left(\sum_{i=1}^N \hat{\mathbf{D}}_i' \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{S}}_i \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right) \left(\sum_{i=1}^N \hat{\mathbf{D}}_i' \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right)^{-1}$$

where $\hat{\mathbf{S}}_i = (Y_i - \hat{\mu}_i)(Y_i - \hat{\mu}_i)'$.

Inference (aka, magic!)

- $\hat{\Sigma}_{\text{Model}}$
 - Requires that $\mathbf{R}_i(\alpha)$ be “correct” for consistency.
 - Is slightly more efficient than $\hat{\Sigma}_{\text{Robust}}$ if so.
- $\hat{\Sigma}_{\text{Robust}}$
 - Is consistent *even if* $\mathbf{R}_i(\alpha)$ is misspecified.
 - Is slightly less efficient than $\hat{\Sigma}_{\text{Model}}$ if $\mathbf{R}_i(\alpha)$ is correct.

Moral: Use $\hat{\Sigma}_{\text{Robust}}$.

GEEs:

- Are a straightforward variation on GLMs, and so
- Can be applied to a range of data types (continuous, binary, count, proportions, etc.),
- Yield robustly consistent point estimates of β s,
- Account for within-unit correlation in an informed way, but also
- Yield consistent inferences even if the correlation is misspecified.

Practical Issues: Model Interpretation

- In general, GEEs = GLMs.
- GEEs are *marginal* models, so:
 - $\hat{\beta}$ s have an interpretation as average / total effects.
 - Estimates / effect sizes generally be smaller than conditional (e.g. fixed/random) effects models.
 - E.g., for logit, $\hat{\beta}_M \approx \frac{\hat{\beta}_C}{\sqrt{1+0.35\sigma_\eta^2}}$, where $\sigma_\eta^2 > 0$ is the variance of the unit effects.
 - See (e.g.) [Gardiner et al. \(2009\)](#) or [Koper and Manseau \(2009\)](#) for expositions.

Practical Issues: Specifying $\mathbf{R}_i(\alpha)$

- Has been called “more art than science.”
- Pointers:
 - Choose based on *substance* of the problem.
 - Remember that $\mathbf{R}_i(\alpha)$ is conditional on $\mathbf{X}$, $\hat{\beta}$.
 - Consider unstructured when T is small and N large.
 - Try different ones, and compare.
- In general, it shouldn't matter terribly much...

Software	Command(s)/Package(s)
R	<code>gee / geepack / geeM / multgeeB / glmtoolbox / repolr</code>
Stata	<code>xtgee / xtlogit / xtprobit / xtpois / etc.</code>
SAS	<code>genmod (w/ repeated)</code>

- Generally follow GLMs (specify “family” + “link”)
- Certain combinations not possible/recommended
- Estimation: Fisher scoring, MLE, etc. (MCMC?)

From the geepack manual:

Warning

Use "unstructured" correlation structure only with great care. (It may cause R to crash).

Civil War Redux... GEE: Independence

```
> GEE.ind<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+
+               log(GDPPerCapita)+GDPPerCapGrowth+PostColdWar+POLITY+
+               POLITYSquared,data=DF,id=IS03,family="binomial",
+               corstr="independence")
```

```
> summary(GEE.ind)
```

Call:

```
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
        UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar +
        POLITY + POLITYSquared, family = "binomial", data = DF, id = IS03,
        corstr = "independence")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-0.2019	1.9351	0.01	0.91691
log(LandArea)	-0.1159	0.1225	0.89	0.34430
log(PopMillions)	0.7000	0.1379	25.76	0.00000039 ***
UrbanPopulation	0.0144	0.0127	1.27	0.25910
log(GDPPerCapita)	-0.4434	0.2197	4.07	0.04362 *
GDPPerCapGrowth	-0.0312	0.0121	6.69	0.00969 **
PostColdWar	-0.4764	0.2599	3.36	0.06686 .
POLITY	0.7283	0.2224	10.72	0.00106 **
POLITYSquared	-0.0695	0.0201	11.90	0.00056 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = independence

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	0.838	0.406

Number of clusters: 160 Maximum cluster size: 60

GEE: Exchangeable

```
> GEE.exc<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+GDPPerCapGrowth+  
+ PostColdWar+POLITY+POLITYSquared,data=DF,id=ISO3,family="binomial",corstr="exchangeable")
```

```
> summary(GEE.exc)
```

Call:

```
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +  
UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar +  
POLITY + POLITYSquared, family = "binomial", data = DF, id = ISO3,  
corstr = "exchangeable")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-3.18808	2.49891	1.63	0.20203
log(LandArea)	0.12732	0.21455	0.35	0.55289
log(PopMillions)	0.47690	0.19054	6.26	0.01232 *
UrbanPopulation	0.00152	0.01162	0.02	0.89618
log(GDPPerCapita)	-0.24778	0.18644	1.77	0.18386
GDPPerCapGrowth	-0.03015	0.00831	13.16	0.00029 ***
PostColdWar	-0.09081	0.24218	0.14	0.70770
POLITY	0.50247	0.17067	8.67	0.00324 **
POLITYSquared	-0.05204	0.01678	9.62	0.00192 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = exchangeable

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	0.855	1.14

Link = identity

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.476	0.374

Number of clusters: 160 Maximum cluster size: 60

GEE: AR(1)

```
> GEE.ar1<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+GDPPerCapGrowth+
+               PostColdWar+POLITY+POLITYSquared,data=DF,id=ISO3,family="binomial",corstr="ar1")

> summary(GEE.ar1)

Call:
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
        UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + PostColdWar +
        POLITY + POLITYSquared, family = "binomial", data = DF, id = ISO3,
        corstr = "ar1")

Coefficients:
              Estimate Std.err Wald Pr(>|W|)
(Intercept)    4.80931   4.99165  0.93   0.335
log(LandArea)  -0.48792   0.45306  1.16   0.282
log(PopMillions) 0.50478   0.38293  1.74   0.187
UrbanPopulation -0.06212   0.07856  0.63   0.429
log(GDPPerCapita) -0.03392   0.61078  0.00   0.956
GDPPerCapGrowth -0.01406   0.00835  2.83   0.092
PostColdWar      0.16271   0.39227  0.17   0.678
POLITY           0.14953   0.18615  0.65   0.422
POLITYSquared    -0.01229   0.01776  0.48   0.489
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = ar1
Estimated Scale Parameters:

              Estimate Std.err
(Intercept)    5.2      198
Link = identity

Estimated Correlation Parameters:
              Estimate Std.err
alpha        0.962    2.02
Number of clusters: 160 Maximum cluster size: 60
```

GEE: Unstructured (2013-2017)

```
> GEE.unstr<-geeglm(CivilWar~log(LandArea)+log(PopMillions)+UrbanPopulation+log(GDPPerCapita)+GDPPerCapGrowth+
+ POLITY+POLITYSquared,data=DF5,id=IS03,family="binomial",corstr="unstructured")
> summary(GEE.unstr)
```

Call:

```
geeglm(formula = CivilWar ~ log(LandArea) + log(PopMillions) +
  UrbanPopulation + log(GDPPerCapita) + GDPPerCapGrowth + POLITY +
  POLITYSquared, family = "binomial", data = DF5, id = IS03,
  corstr = "unstructured")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-2.7230	3.1314	0.76	0.38453
log(LandArea)	0.0213	0.1875	0.01	0.90955
log(PopMillions)	0.8530	0.2347	13.21	0.00028 ***
UrbanPopulation	0.0307	0.0154	3.99	0.04578 *
log(GDPPerCapita)	-0.5749	0.2870	4.01	0.04514 *
GDPPerCapGrowth	0.0314	0.0286	1.20	0.27340
POLITY	0.5687	0.4244	1.80	0.18023
POLITYSquared	-0.0598	0.0378	2.50	0.11367

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = unstructured

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	0.673	0.431
Link = identity		

Estimated Correlation Parameters:

	Estimate	Std.err
alpha.1:2	0.582	0.342
alpha.1:3	0.535	0.338
alpha.1:4	0.436	0.301
alpha.1:5	0.271	0.197
alpha.2:3	0.863	0.495
alpha.2:4	0.373	0.277
alpha.2:5	0.385	0.266
alpha.3:4	0.477	0.346
alpha.3:5	0.599	0.400
alpha.4:5	0.430	0.317

Number of clusters: 157 Maximum cluster size: 5

GEE Model Comparison (2013-2017)

Table 1: GEE Models of Civil War Onset

	Independence	Exchangeable	AR(1)	Unstructured
(Intercept)	-3.715 (4.025)	-2.423 (3.445)	-4.451 (3.711)	-2.723 (3.131)
ln(Land Area)	0.024 (0.193)	0.039 (0.204)	0.152 (0.208)	0.021 (0.187)
ln(Population)	0.859*** (0.237)	0.818*** (0.237)	0.731** (0.251)	0.853*** (0.235)
Urban Population	0.025 (0.018)	0.027 (0.017)	0.021 (0.016)	0.031* (0.015)
ln(GDP Per Capita)	-0.473 (0.340)	-0.579* (0.289)	-0.444 (0.297)	-0.575* (0.287)
GDP Growth	-0.018 (0.035)	-0.007 (0.023)	0.027 (0.036)	0.031 (0.029)
POLITY	0.886 (0.553)	0.504 (0.429)	0.528 (0.516)	0.569 (0.424)
POLITY Squared	-0.090+ (0.047)	-0.053 (0.037)	-0.056 (0.046)	-0.060 (0.038)
Num.Obs.	781	781	781	781

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

GEEs are:

- Robust
- Flexible
- Extensible beyond panel/TSCS context

A few useful resources:

- The GEE [Wikipedia page](#)
- [Liang and Zeger \(1986\)](#)
- [Vanegas et al. \(2023\): “Generalized Estimating Equations Using the New R Package glmtoolbox.”](#)
- Textbooks, etc.:
 - [Diggle et al. \(2002\)](#)
 - [Zeigler \(2011\)](#)
 - [Hardin and Hilbe \(2013\)](#)

Thank you!